CSE 340: Computer Architecture

[Spring 2025]
Quiz - 1 :: Set - A

Time: 15 min **Marks: 15**
Name: ______________________________
ID:______________________ **Section:** ______________________

Three processors, P1, P2, and P3, execute the same instruction mix but with different clock rates and CPI values:

Processor	Clock rate	CPI
P1	3.1 GHz	1.4
P2	3.9 GHz	1.2
P3	4.2 GHz	1.6

A program with 2.4×10^6 instructions is executed.

1. Compute the execution time for each processor
2. Which processor has the highest throughput (instructions per second)?
3. If P2 improves CPI by 15% but reduces clock rate by 11%, does it outperform P1?

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Quiz - 1 :: Set - B

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A processor executes four instruction types (A, B, C, and D) with the following CPIs:

Instruction Type	CPI	Occurrence (%)
A	7	26%
B	2	37%
C	4	19%
D	3	18%

The processor runs at 3.2 GHz, executing 1.9 x 10^6 total instructions.

1. Compute the average CPI for this workload.
2. Calculate the execution time in milliseconds.
3. If the CPI of instruction A is reduced by 25%, how much will the execution time improve?